WEEK 1 · DAY 1 · INSTRUCTOR NOTES

Colab Bootcamp + Your First Time Series

For the trainer only.

The one idea for today: *a time series is a table where the order of the rows carries meaning.* Everything else (Colab, pandas, `.asfreq`) is the tooling to work with that idea.

At a glance

Length	~2 hours (it can stretch to 2.5 if Colab sign-in is slow)
Participants leave able to	open and save a Colab notebook, run cells in order, recover from a restart, load a CSV as a proper time series, find gaps, name the 4 components
Covers (original)	Week 3 Code Book: *Setup*, the `profile()` helper, *Conventions used in every file*, **B1 Ka-band availability daily**, **B2 monthly**; Week 3 quiz topic *structure of time-series data*
Dataset	`Time-series/week-03/data/ka_band_availability_daily.csv` (+ `noc_tickets_daily.csv` for the bonus)
Files you need open	animated slides (`1-slides/web-animated/`), your **solution** notebook, the rule book PDF

The day before: checklist

☐ Open the exercise link yourself in a fresh Chrome window and run STEP 0. You should see ✅ `Ready!`

☐ Run your **solution** notebook top to bottom: every check should be ✅ (it takes about 20 seconds).

☐ Ask participants to bring a laptop with **Chrome** and to know their **Google account password**. A personal Gmail is fine.

☐ Some work Google accounts block Colab. If so, participants use a personal Gmail for the course.

☐ Test the projector with the animated slides: press **F** for full screen and **N** to see these notes on your laptop.

☐ Share the rule book PDF and the exercise link in the class group chat (WhatsApp/Teams).

Run of show (2 hours)

Time	Block	What happens	Slides
0:00	**Hook**	NEPA light diary: "a time + a value = a time series"	1–3
0:10	**Colab bootcamp**	What Colab is, the 5 parts of the screen, the 3 golden rules. **Everyone saves a copy in Drive NOW.**	4–7

Time	Block	What happens	Slides
0:25	**Live demo**	Restart → NameError → fix it. The AI-button data rule.	8–9
0:35	**What is a time series**	Order matters (shuffle demo), anatomy, NigComSat-1R, today's data	10–14
0:50	**The recipe**	parse_dates → index_col → asfreq; the 5 inspection questions; UTC vs WAT; 4 characters	15–19
1:00	**Hands-on**	Exercise Part A (15 min) then Part B (45 min). Walk the room.	20
1:45	**Wrap-up**	Scoreboards up, recap, 2 volunteers explain "order matters" and "why asfreq"	21
1:55	**Exit**	Share the glossary PDF as revision. Tease Day 2.	–

Explain it like this (in plain English)

- **Colab** = Python on Google's computer. The laptop is just a screen. Nothing to install.
- **Time series** = readings in time order. If you can shuffle the rows without losing anything, it's *not* a time series (a shopping list). If shuffling destroys the story, it *is* (the Ka-band chart).
- `parse_dates` : pandas reads dates from a CSV as text until you tell it otherwise. Text can't do "give me August".
- `index_col` : makes the dates the row labels, so you can say `ka.loc["2024-08"]` .
- `.asfreq("D")` : the most important line. If an export is missing 3 days, `len()` still looks normal. `.asfreq` creates a slot for every day, so the missing ones show up as `NaN` and you can count them.
- **UTC vs WAT**: satellite telemetry is UTC, Lagos clocks are UTC + 1. Join two tables in different time zones and every event moves by an hour.
- **The 4 characters**: *trend* (direction), *seasonality* (fixed calendar), *cycle* (no fixed length), *noise* (ignore it). The seasonality-vs-cycle difference trips people up: the calendar decides seasonality; nobody can schedule a cycle.

Slide talking points

#	Say this	Ask this
3	"You've been collecting time series all your life."	"How many hours of light did you get yesterday?"
5	"Even a small laptop is fine; the work happens in Google's data centre."	–
7	Stop and make everyone do Rule 1. Walk the room and check.	"Whose notebook title now starts with *Copy of*?"
8	Demo the restart live. Make the NameError happen on purpose.	"What does NameError mean?" → "Colab forgot."
9	The training data is synthetic and safe. Real data is restricted.	–

#	Say this	Ask this
11	Press next to shuffle. Pause. Let them see the story vanish.	"Same numbers. What changed?"
13	Launched Dec 2011, replaced NigComSat-1 (lost in 2008 after a solar-array failure). 28 transponders, 4 bands.	–
14	Point at the dips: every rainy season. Ka-band hates heavy rain, like DStv.	"Which month do you think is worst?" (don't answer: they find it in B8)
16	Count the boxes in the top row: 11. Then the bottom: 14.	"Which one would you rather trust?"
19	"38 vs 41 tickets is the same week twice. Don't call a meeting."	–

Answer key

Task	Answer	Common mistake → what to say
A2	`24 * 365` → 8760	–
A3	Restart → `NameError` → run STEP 0 + A2	They forget STEP 0 after restarting → "STEP 0 after every restart."
B1	`"text"`	Typing `object` or `str` is also accepted
B2	`"timestamp"`	No quotes → `NameError` → "text needs quotes"
B3	`len(ka)` → 1,096	–
B4	`ka.index.min()` / `ka.index.max()` → 2023-01-01 / 2025-12-31	Using `ka["timestamp"]` → it's the index now, not a column
B5	`FREQ = "D"` → 3 missing days	`FREQ = D` without quotes
B6	`ka["availability_pct"].idxmin()` → **2023-07-25** (95.71%)	`.min()` gives the value, not the date. `idxmin` gives the date.
B7	Any sentence that states a finding	"Availability" alone is a label, not a finding
B8	`"August"`	–
B9	seasonality, trend, noise, cycle	Mixing up seasonality and cycle (fixed calendar vs not)
B10	`22`	Answering 20 (subtracting instead of adding)
🌶️	`"Sunday"` (Saturday is close behind)	Forgetting `parse_dates` / `index_col` → `.day_name()` fails

Quiz link

The original Week 3 quiz (`Time-series/MCQ/week-03-mcq.csv`), topic **"structure of time-series data"**, questions 1–6, is exactly today's content: order matters, the 4 components, noise vs change, seasonality vs cycle, `.asfreq` , time zones. Use them as a 5-minute warm-up at the start of Day 2.

Troubleshooting

Problem	Fix
"Colab is blocked / disabled for your organisation"	Use a personal Gmail account
STEP 0 fails with `HTTP Error` / `URLError`	Rare: GitHub is briefly unreachable from Google. Wait a minute and re-run. Fallback: download the CSV from the repo, drag it into the 📁 Files panel, and STEP 0 will use the uploaded copy
"Runtime disconnected"	Idle too long. Click *Reconnect*, then run STEP 0 and the cells above
Someone's checks pass but their chart is empty	They ran the plot before loading `ka` . Restart and run all
A participant finishes in 20 minutes	Send them to the 🌶️ bonus, then ask them to help a neighbour

Notes on the original material

- The original code book loads data with `DATA = Path("data")` , which **doesn't work in Colab**. Our `data_path()` downloads each file from the original repo instead.
- The original uses `.asfreq("H")` , which newer pandas prints warnings about. We use `"h"` (hourly) and `"D"` (daily).
- Facts about NigComSat-1R used on the slides are in `toolkit/nigcomsat-fact-sheet.md` , with sources.